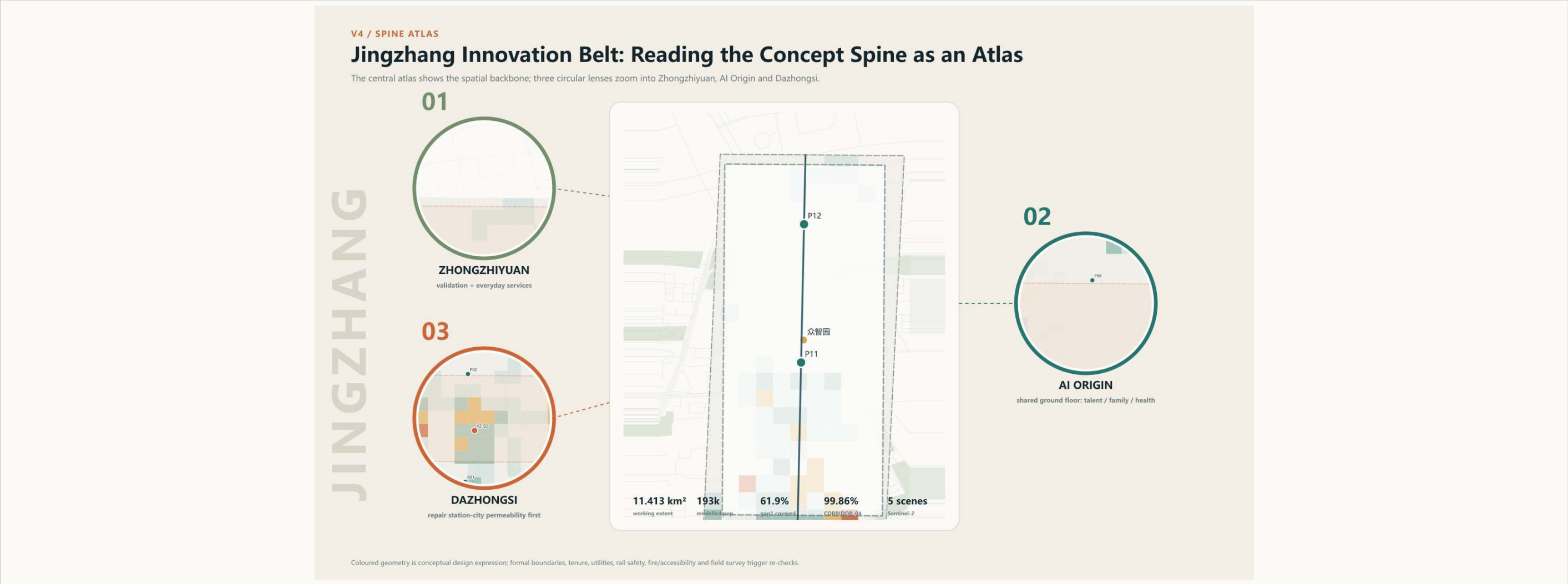


CENTENNIAL JINGZHANG AI INNOVATION BELT

FROM URBAN DIAGNOSIS TO AUDITABLE DESIGN



11.41 km²
provisional overall-design base

192.6k
WorldPop modelled population

17.9%
population with ≥ 1 service gap

61.9%
V2 consensus gap1 coverage

Existing Evidence and Overall Spatial Structure

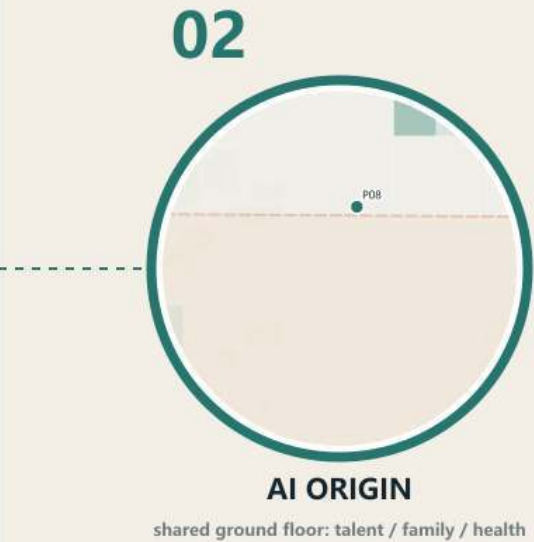
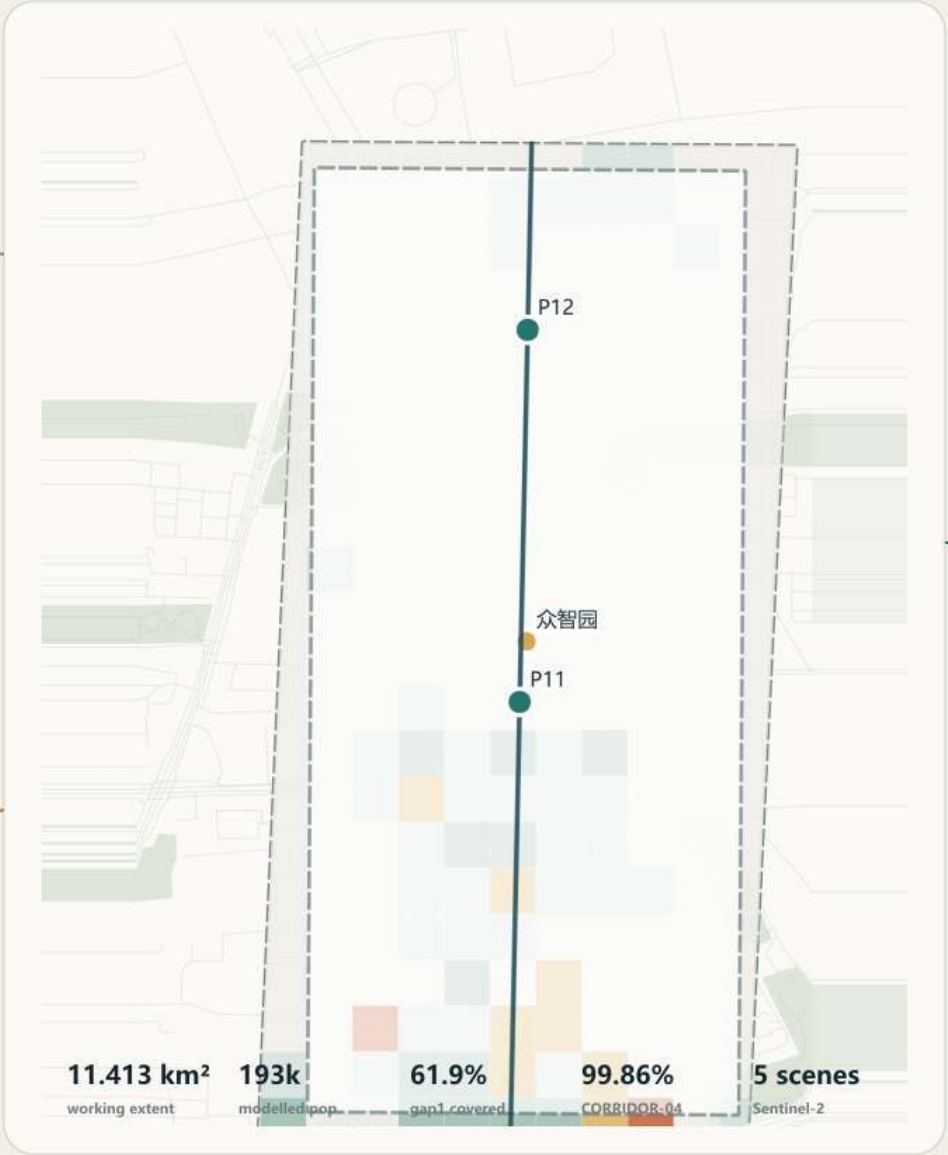
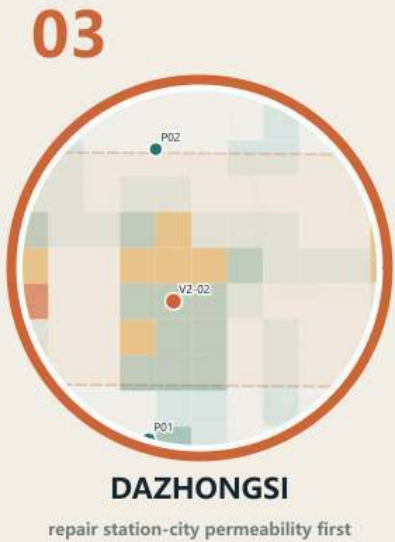
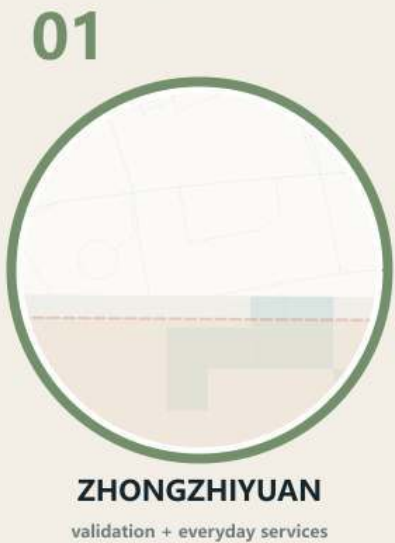
One evidence base constrains both diagnosis and concept design

V4 / SPINE ATLAS

Jingzhang Innovation Belt: Reading the Concept Spine as an Atlas

The central atlas shows the spatial backbone; three circular lenses zoom into Zhongzhiyuan, AI Origin and Dazhongsi.

JINGZHANG



WorldPop modelled population, OSM walk network, robust opportunity and promoted V2 geometry are shown together. Coloured geometry is conceptual design expression; formal boundaries, tenure, utilities, rail safety, fire/accessibility and field survey trigger re-checks.

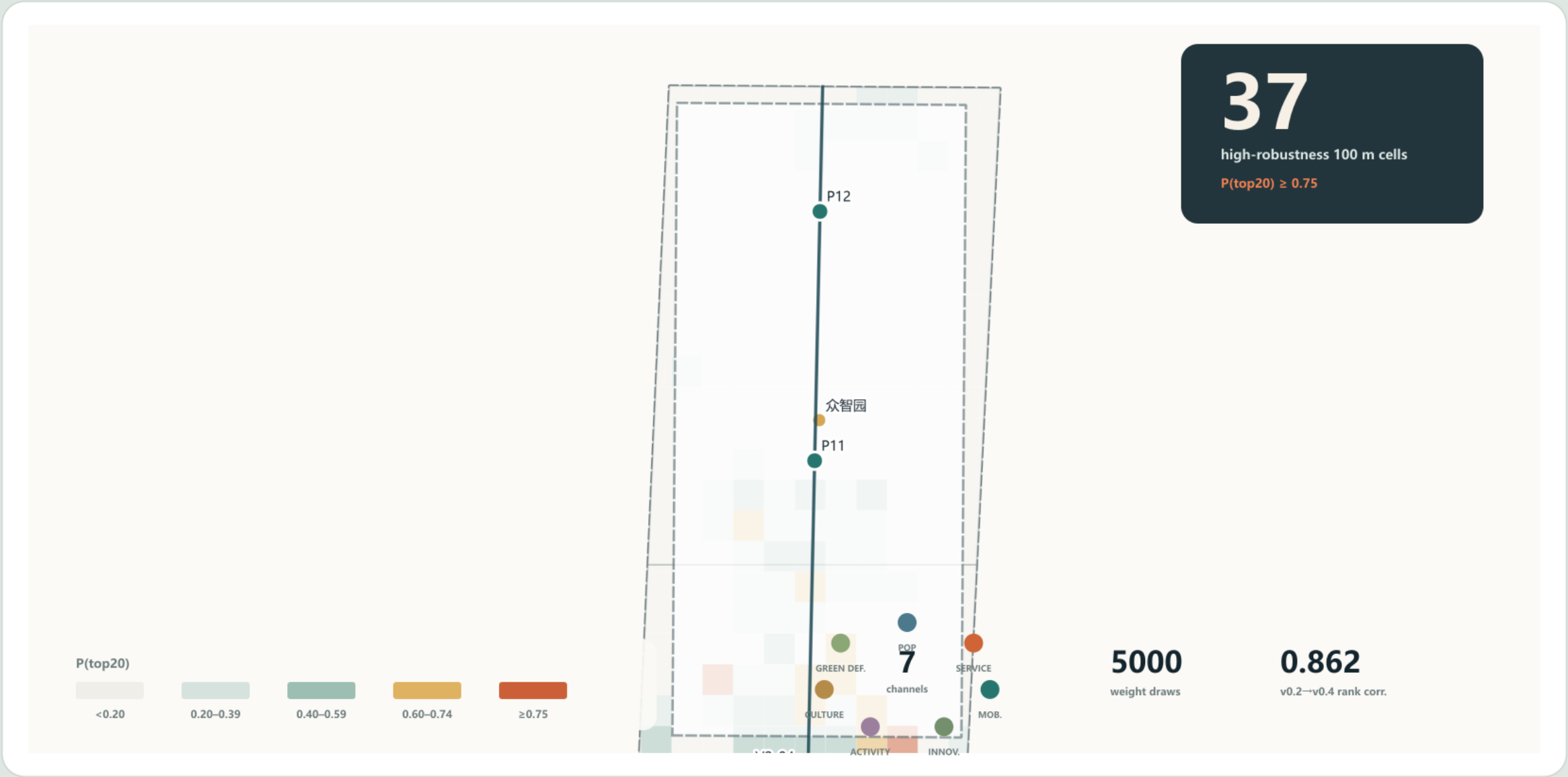
Robust Opportunity without a Single Arbitrary Weighting

Seven percentile-normalized dimensions + 5,000 weight samples

V4 / EVIDENCE LANDSCAPE

Robust Opportunity Is a Probability Landscape, Not a Single Score

A 100 m cell is robust only when it repeatedly enters the top 20% across 5,000 plausible preference draws.



In Sentinel v0.4, 37 100 m cells have $P(\text{top20}) \geq 0.75$; the green dimension uses the five-date Sentinel-2 summer vegetation density. V0.4 replaces the old QSM green proxy with a five-date Sentinel-2 vegetation deficit and uses tie-aware ranks. Warm colour expresses robust priority, not statutory development intensity.

The Three Key Areas Require Different Diagnoses

Population-weighted mobility, health and service gaps define the first move

03 / THREE URBAN CASES

THREE KEY AREAS, THREE DIFFERENT FIRST MOVES

Population-weighted mobility, health and service gaps define three distinct first-stage spatial prototypes.

ZHONGZHIYUAN

01

VALIDATION + EVERYDAY-SERVICE COMPLETION



FIRST-STAGE PROJECT

The ~1.5 ha test field is a research hypothesis and must be checked together with real everyday-service deficits.

Concept visualization / not factual evidence

AI ORIGIN COMMUNITY

02

SHARED GROUND FLOOR: TALENT / FAMILY / HEALTH



FIRST-STAGE PROJECT

Mobility is comparatively mature; prioritize learning, talent/family support and health referral.

Concept visualization / not factual evidence

DAZHONGSI

03

STATION-CITY CONNECTIVITY FIRST



FIRST-STAGE PROJECT

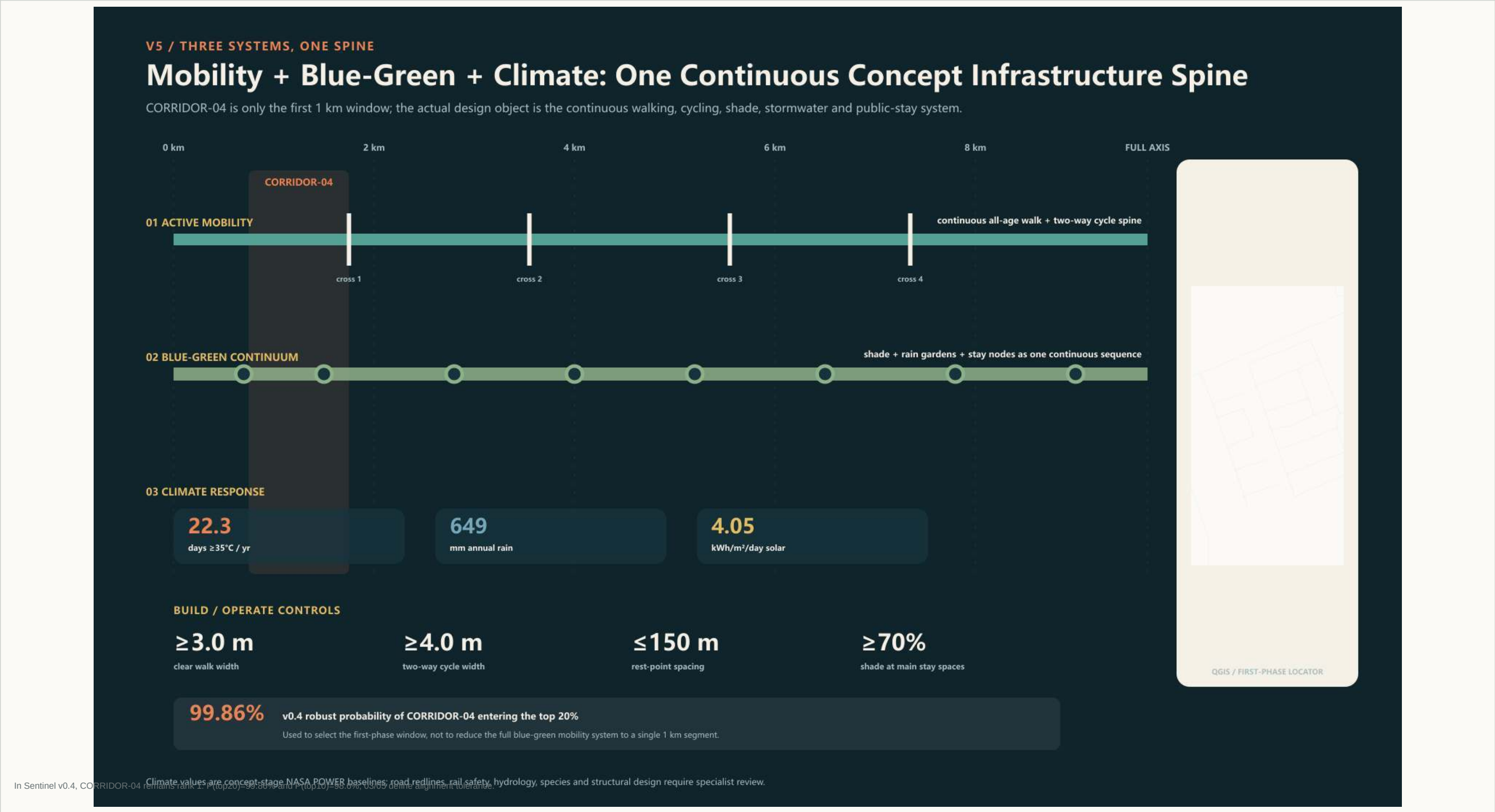
Low rail access, high service gaps and the largest detour proxy put walking/cycling connectivity first.

Concept visualization / not factual evidence

Zhongzhiyuan completes services, AI Origin focuses talent/family support, and Dazhongsi repairs station-city connectivity first.

Why the One-kilometre Pilot Starts on the Dazhongsi Side

36 sliding windows + 5,000 weight samples



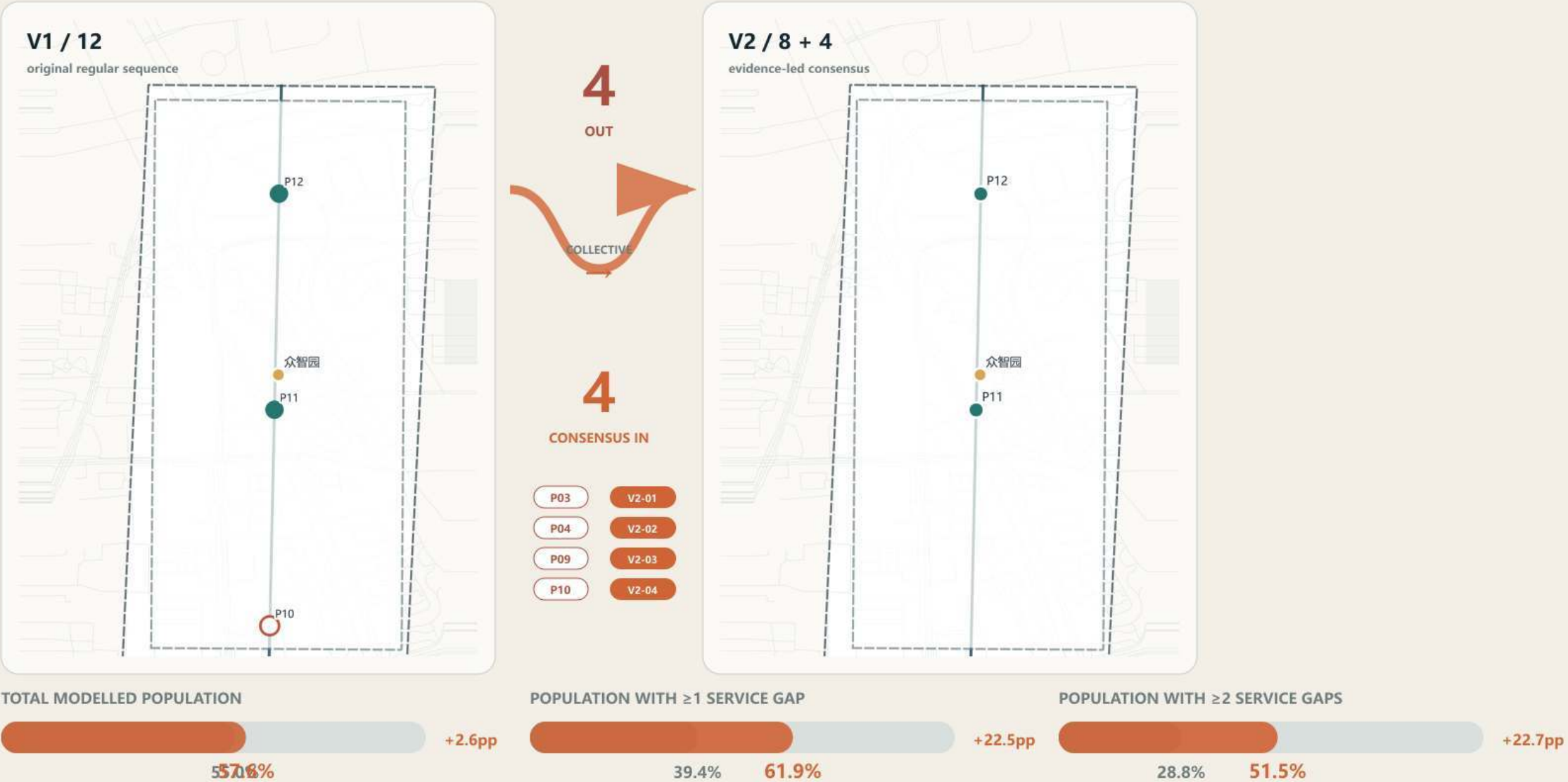
Twelve Nodes: From Priori Locations to Multi-objective Consensus

Retain eight centers and replace four low-performing slots with consensus sites

V4 / BEFORE → AFTER

Same 12 Nodes, Better Coverage: What Changed in V2?

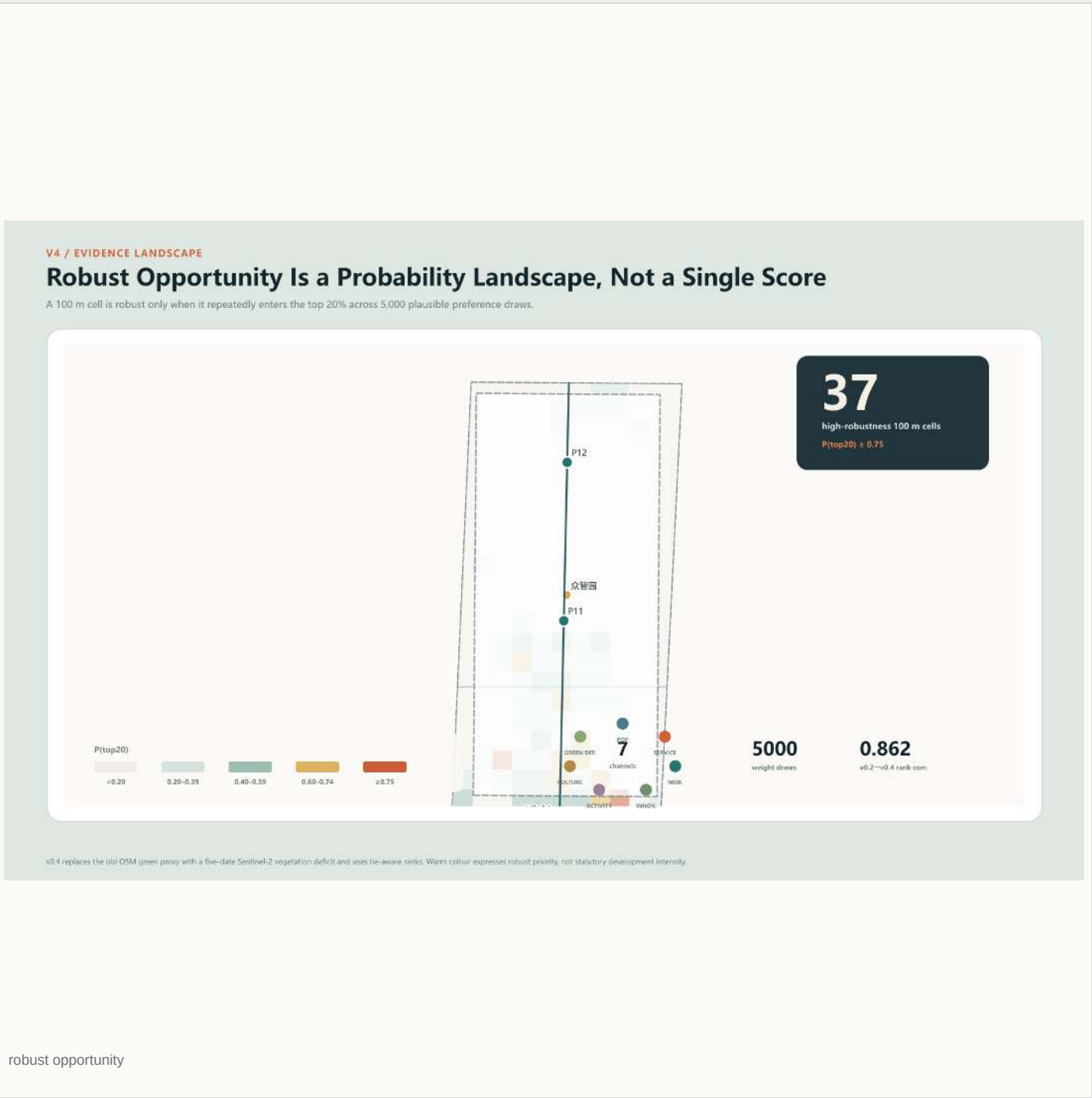
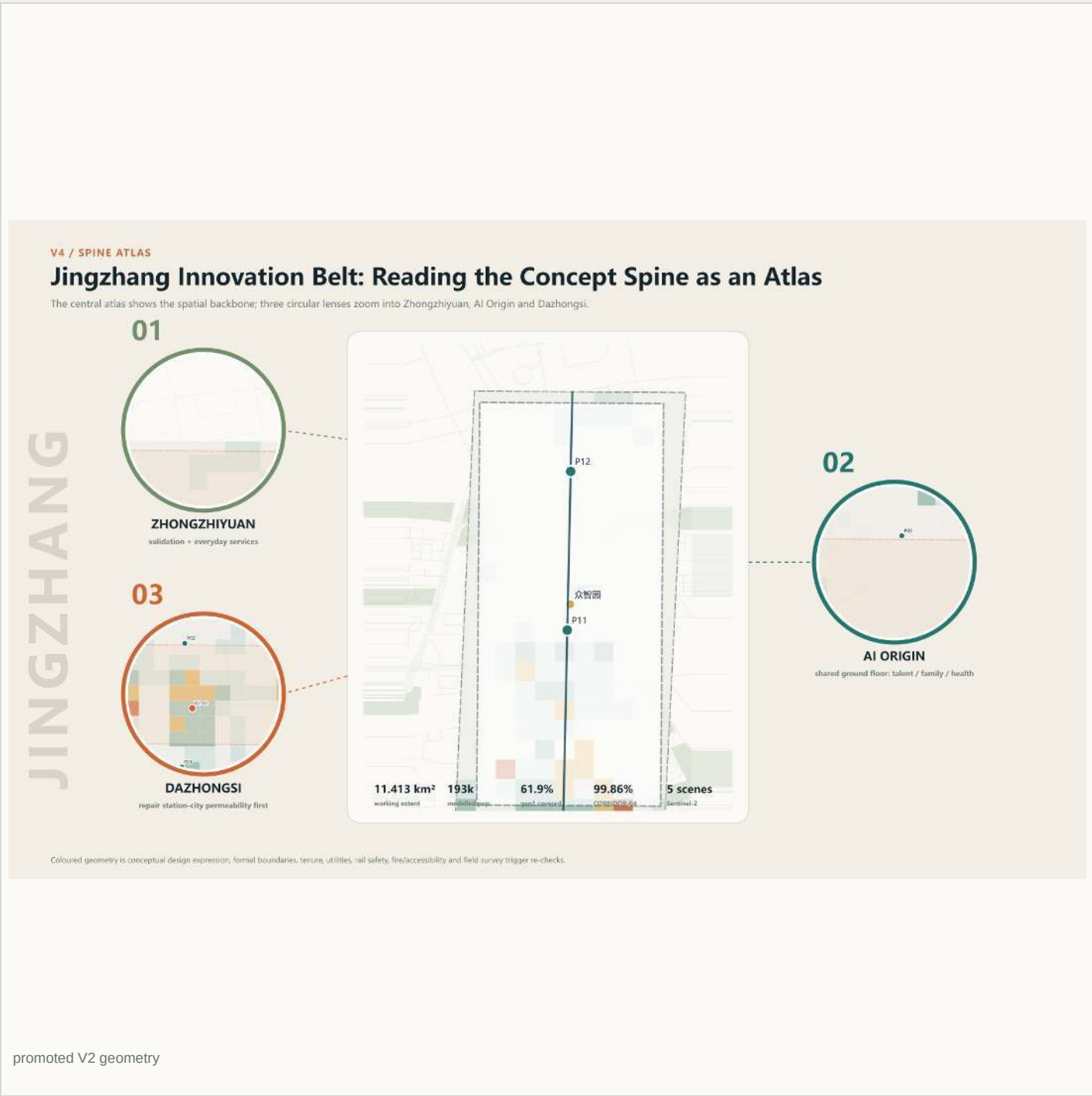
Eight centers stay; PUBLIC-03/04/09/10 leave low-deficit positions and four multi-objective consensus replacements move toward measured service gaps.



The full Sentinel v0.4 rerun retains the current four consensus replacements; twelve-node gap1 coverage stays at 61.9% while the alternative gains eight opportunity intensity at the cost of population and gap1 coverage.

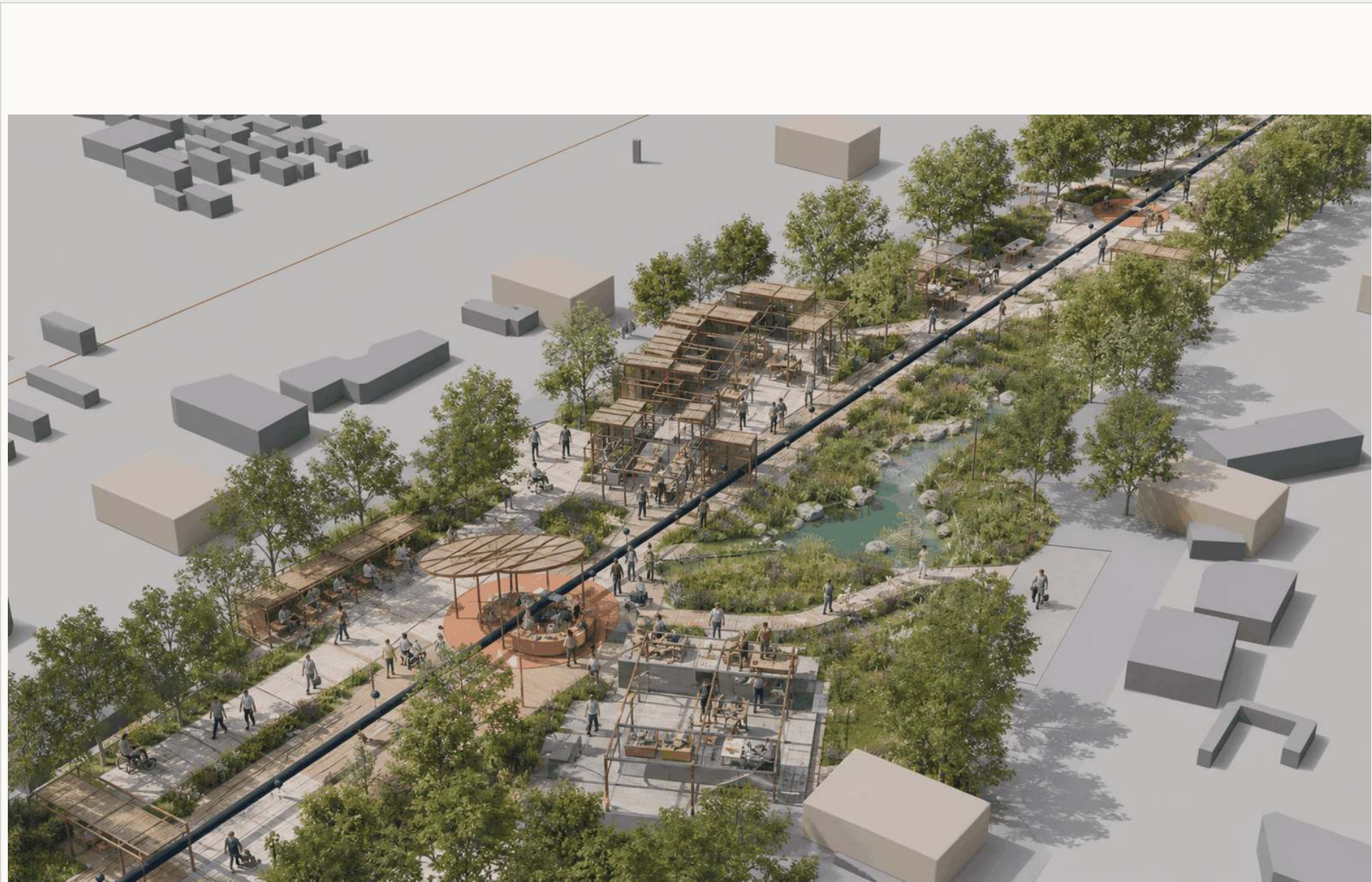
Overall Structure: Opportunity, Nodes and Key Areas Work Together

Concept design remains subject to formal redlines, tenure, utilities and field checks



Zhongzhiyuan | Couple Validation with Everyday-service Completion

The ~1.5 ha test field remains a research hypothesis but must be re-sited and checked against real service deficits.



Concept visualization / not factual evidence

22.6k
modelled population

53.0%
rail within 10 min

79.5%
health within 15 min

36.8%
gap in ≥1 service

1.94×
pedestrian detour proxy

AI Origin | Shift the Shared Ground Floor toward Talent and Life Support

Mobility is relatively strong; the first stage should emphasize talent, family, health referral, learning/collaboration and open innovation.



Concept visualization / not factual evidence

14.9k
modelled population

86.7%
rail within 10 min

63.5%
health within 15 min

36.5%
gap in ≥ 1 service

1.71x
pedestrian detour proxy

Dazhongsi | Repair Station-city Connectivity before Expanding the Public Hall

Low 10-minute rail coverage, high health deficit and the largest detour proxy place connectivity ahead of the 2.4 ha public-hall expansion.



Concept visualization / not factual evidence

18.0k
modelled population

25.3%
rail within 10 min

49.6%
health within 15 min

53.6%
gap in ≥ 1 service

2.78x
pedestrian detour proxy

320 m² Service Node: Small, Demountable and Staffed

Total concept footprint is 3,840 m²; module size is not an approved construction area

10 / SERVICE NODE GUIDE

Public-service Node Delivery and Operations Guide

A 320 m² prototype defines site, program, staffing, utilities, acceptance and exit requirements



NODE AXONOMETRIC INTENT | 20 m × 16 m | continuous accessible loop

NODE DELIVERY + OPERATIONS

250–400 m² program range | 320 m² prototype

01 FOOTPRINT 320 m²

20 m × 16 m prototype with demountable components

ACCEPT site and built-area check

02 HUMAN SERVICE 24–36 m²

counter, waiting, storage and staffed position

ACCEPT hours and staffing roster

03 MICROCLIMATE ≥70%

continuous shade across principal stay areas

ACCEPT summer solar study + field audit

04 RESPONSE 60 s / 24 h

human acknowledgement and routine fault logging

ACCEPT monthly response + work-order log

HANDOVER + EXIT MANAGEMENT

- joint utilities, fire, access and site-permit review
- facility list, duty register and monthly inspection record
- remove equipment if operations lapse; retain basic civic amenities

The four new sites still require official-facility overlap, tenure, utilities, fire, accessibility and field checks. generated-service-node-axonometric.jpg

node delivery integrates utilities, fire review, accessibility and operating maintenance

Pilot Section: Translate Accessibility into Design Controls

clear walk ≥3.0 m; two-way cycle ≥4.0 m; effective rest spacing ≤150 m

09 / TYPICAL SECTION

Typical-section Controls for the Jingzhang Pilot

A 12-m typical section addresses walking gaps, mixed traffic, limited rest and stormwater drainage



SECTION CONTROLS + ACCEPTANCE

12.0 m total | four continuous functional bands

01	WALKING	≥3.0 m
	continuous clear width, tactile and kerb-ramp links	
	ACCEPT field width survey	
02	STORMWATER	≥2.5 m
	complete inflow, overflow and drainage route	
	ACCEPT hydraulic check + flow test	
03	CYCLING	≥4.0 m
	two-way operation with speed control at crossings	
	ACCEPT marking width + conflict review	
04	SERVICE EDGE	≥2.5 m
	seating, shade and staffed wayfinding	
	ACCEPT facility list + access audit	

DELIVERY INTERFACES

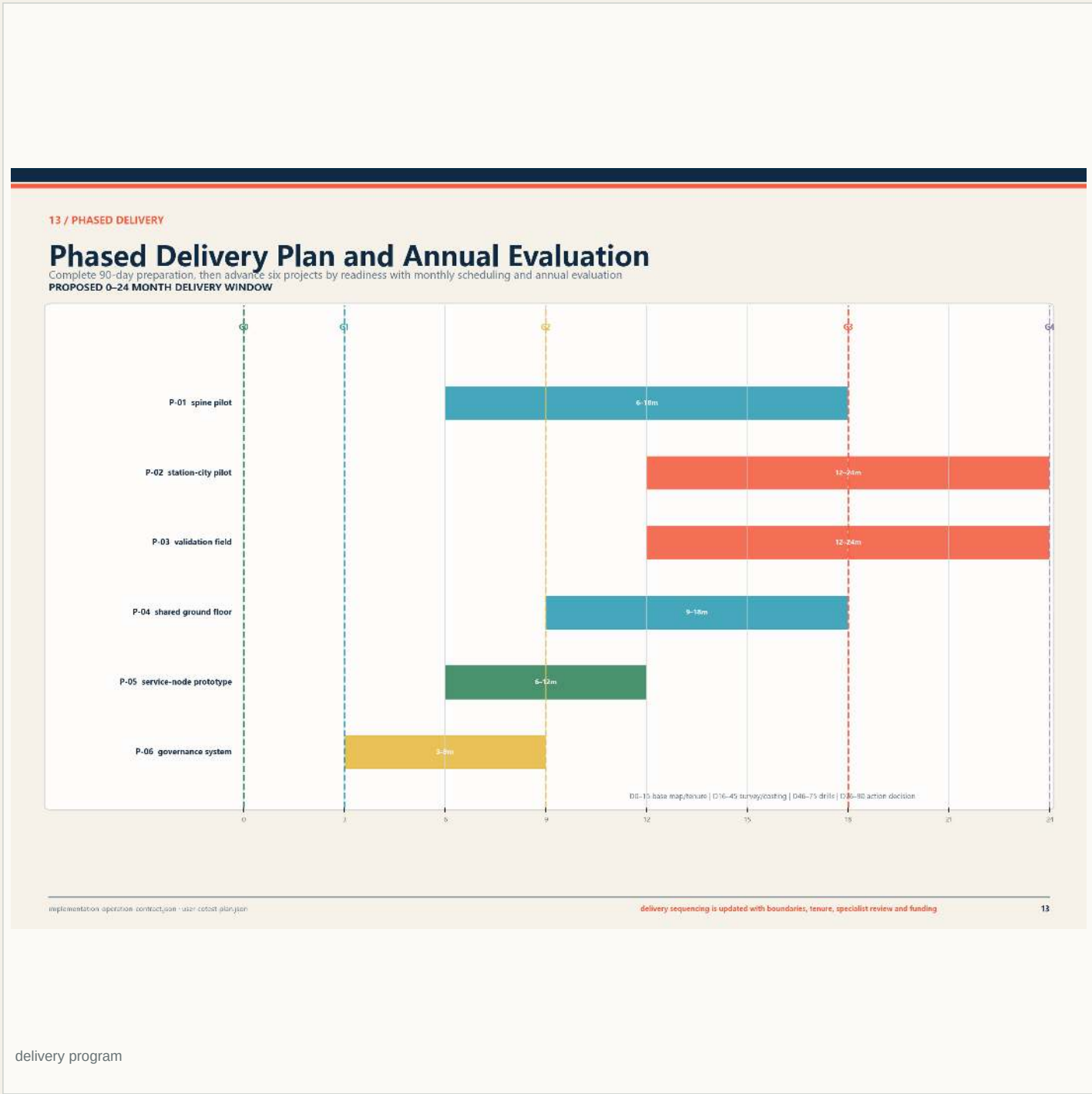
- coordinate rail protection and maintenance access
- integrate street limits, utilities and existing trees
- retain physical wayfinding and human service during outages

Controls still require coordination with formal road regulations, rail safety, fire access, utilities and existing trees

section dimensions are developed with rail safety, street limits, utilities and existing trees

0-24 Months: Near-term Packages and Delivery Sequence

Break design moves into reviewable, reversible and annually evaluated delivery packages



Inclusion and Annual Operations: AI Service Requires Human Fallback

The 8-80 equal-service contract spans facilities, information and operations

07 / INCLUSIVE SERVICES

Priority-user Needs and Project Rectification List

Complete priority-user walk-throughs before initiation and record issues, rectification duties and responses

REGISTERED USER ISSUES 0-3

Children and families	3		2		3		
Older people and caregivers	1	3	3				1
Disabled users	3		1		1	2	
Long-term residents			1		3		1
Business and small businesses	1		1	3	3		1
Students and young talent			3	3		1	
Commuters and visitors	3				1		
Facilities service and maintenance staff			1	2		1	3
Neighbourhood parks and public institutions				1	2	1	
	mobility	rest + care	human service	learning + work	delivery burden	data + digital	maintenance + takeover

SURVEY + ACTION ARRANGEMENT

9类

priority-user groups

6项

participation steps

1套

rectification register

Coverage is calculated after population, employment and time-of-day flow baseline; current action to meet group issue lists.

inclusion ledger/guide - user cohort planning

all-age needs are integrated into consultation, design review and operating evaluation

07

inclusion and incidence

18 / ANNUAL OPERATION

Annual Events, Developer Community and Conversion

Four seasons, four working groups and a five-gate pathway share one annual programme

FOUR-SEASON ACTION CALENDAR

SPRING

URBAN PROBLEM SEASON

30 needs + 6 site walks

scenario register

SUMMER

OPEN VALIDATION SEASON

20 prototypes + 8 negative tests

test + connection records

AUTUMN

CIVIC INTELLIGENCE WEEK

12 co-tests + 4 workshops

co-test + open output/ dialogue

WINTER

ANNUAL GOVERNANCE ASSEMBLY

12 services + 6 packages reviewed

continue / correct / reduce / stop list

FOUR DEVELOPER WORKING GROUPS

01 public service + accessibility

02 model safety + evaluation

03 edge devices + robotics

04 urban data + maintenance

ANNUAL: 12 open workdays and 8 reusable outputs

FIVE-GATE CONVERSION PATHWAY

REGISTER

VALIDATE

CO-TEST

PROCURE

SCALE

NEXT GATE: rights clear, cost affordable, owner assigned and correction closed.

Events, site maintenance and staffed service use separate budgets.

annual operation programme/guide - growth handbook/guide

annual receipts are public; missed targets trigger correction, reduction or exit

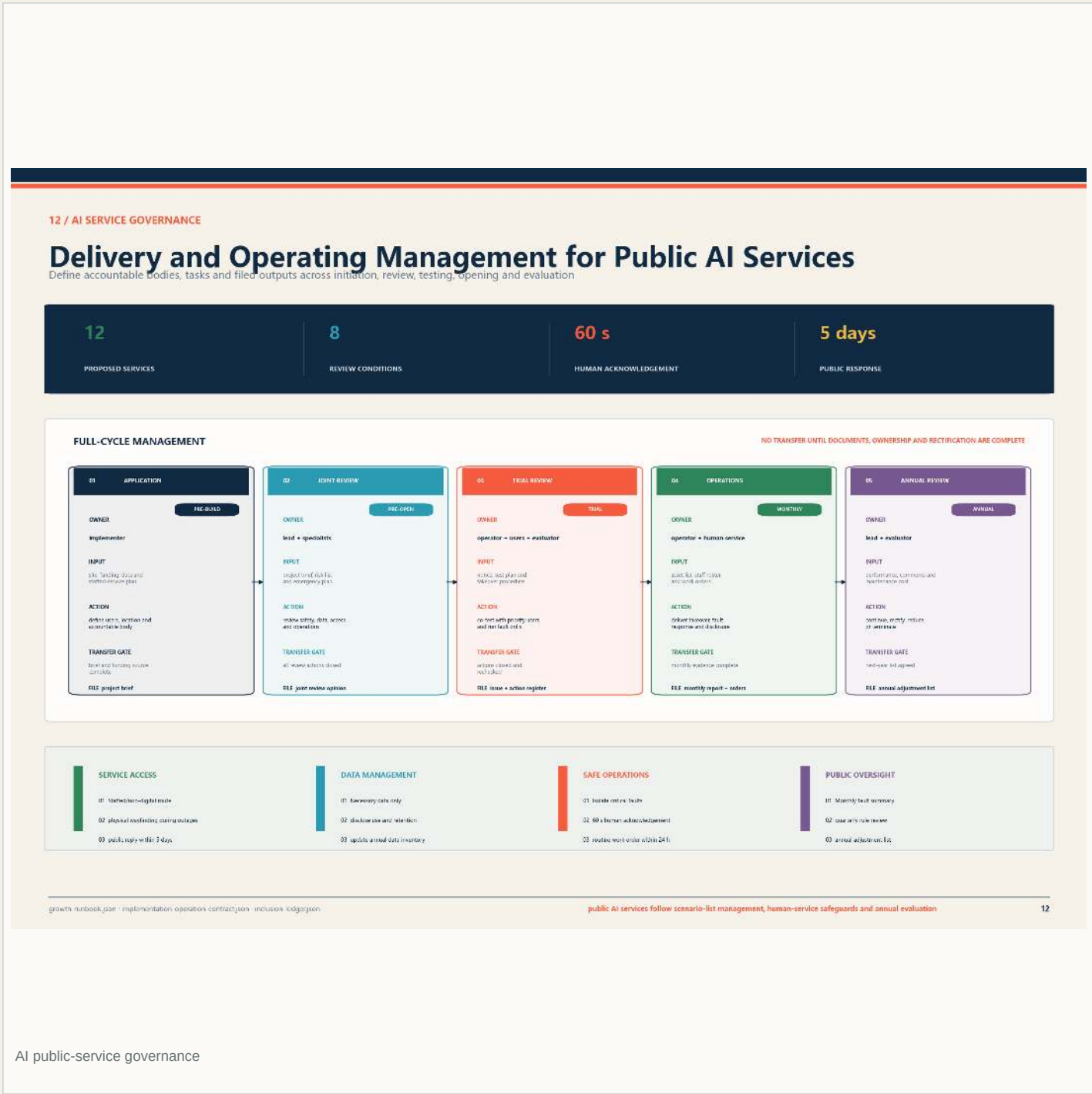
18

annual operating cycle

Human-readable drawings; authoritative data remain in JSON/GeoJSON evidence files

AI Governance and Trial-operation Protocol

Technical launch, public feedback, incident tickets and annual review share one governance loop



Public Identity and Jingzhang Culture: Narrative Serves Everyday Space

Brand, cultural nodes and identity do not replace spatial evidence

14 / PUBLIC IDENTITY

Public-space Identity and Wayfinding Guidelines

A three-level system coordinates corridor gateways, direction signs and service confirmation

GROW WITH JINGZHANG
RAIL GATEWAY - WAYFINDING - HUMAN SERVICE - NIGHT IDENTITY

HIERARCHY + APPLICATION

A continuous, legible and all-age public information system

L1 | CORRIDOR GATEWAY

All street entrances and community entrances, cross identity and overall cohesion

L2 | DIRECTION SIGN

All route directions, stop destinations, distance, approximate costs and interchange

L3 | SERVICE CONFIRMATION

All service nodes, state, hours, human desk, status and feedback route

INFORMATION + MAINTENANCE

- Separate project identity from government systems and maintain a clear information boundary
- Encourage service points, clear human service sign and non-digital access restrictions
- Local media duty, service status and stop information, easy and open support and no redundancy
- Regular monthly media and public information work and 24-hour

image2: public identity.jpg | brand system.json | implementation operation contract.json

public identity is designed, installed and maintained through the delivery program

14

identity system

17 / CULTURE + LANDMARKS

Cultural Narrative, Civic Landmarks and Recognition

Three narratives become everyday civic facilities, contributions receive annual review

THREE CIVIC NARRATIVES

INDEPENDENT ENGINEERING

1909—1949

Rail engineering, survey and construction archive

OPEN INNOVATION

1980—2025

Zhongguangyun research, open source and transfer

CIVIC INTELLIGENCE

2026—

Rights, duties and rules for AI public service

THREE CIVIC LANDMARKS

L-01	ORIGIN MILESTONE COURT AI Origin Community	1909 / 1960 / 2026 archive and courses 1 model/story/set + 1 course / year
L-02	OPEN MODEL GALLERY Zhongguangyun	Models, methods, failure cases and stop records 3D simulation records / year
L-03	CIVIC GOVERNANCE SIGNAL TOWER Daxiongqi Public Hall	Live status for twelve public AI services Live status + annual public report

OPEN CONTRIBUTION ARCHIVE

problem discovery

open tools + data

safety + accessibility

long-term maintenance

AWARD RULE
Public output, co-authorship and verifiable record; sponsorship alone does not qualify

culture: landmarks system.json | brand system.json

landmarks combine access, rest, learning, wayfinding and staffed service

17

heritage / honour / pilgrimage nodes

Human-readable drawings; authoritative data remain in JSON/GeoJSON evidence files

AI Ecosystem and Regional Collaboration: External References for Programming

Cases and regional collaboration inform programming but are not existing-condition spatial data

16 / ECOSYSTEM + BENCHMARK

Benchmark Cases and AI Innovation Ecosystem

Six comparisons set policy choices; four stages and eight factors govern annual delivery

SIX BENCHMARKS: ADOPT / REJECT

Kendall Square

ADOPT: Continuous validation and transfer frontage

REJECT: Public service and R&D collaborations

Paris-Saclay

ADOPT: Synchronise transit and shared facilities

REJECT: Renewal and transacted planning

Kalasatama

ADOPT: Register tests, takeover and annual review

REJECT: Public-service continuity

Punggol Digital District

ADOPT: Share scenarios across park, university and community

REJECT: Translators and transferable interfaces

Barcelona Superblocks

ADOPT: Recover public space through mobility works

REJECT: Traffic assessment before capacity change

Vienna Gender Mainstreaming

ADOPT: Make priority group walks part of initiation

REJECT: Priority-group demand registers

FOUR-STAGE INNOVATION CHAIN

NEEDS

30 needs / year

VALIDATE

20 prototypes / year

CO-TEST

12 co-tests / year

CONVERT

8 conversion plans / year

COMMON RECEIPT: ID, provenance, safety review, correction and conversion result

EIGHT FACTOR RULES

LAND

Existing assets first

INDUSTRY

Five-sector service list

TALENT

Future-to-job pathway

DATA

Five data data register

SPACE

One-third paths to mobility conversion

FINANCE

Capital + 5-year OFEX

COMPUTE

Three-nodes maturing

SCENARIO

Open-call to walk update

ecosystem: policyjoin + six registered background cases

benchmarks inform method; annual targets enter programme control

16

ecosystem benchmark

11 / REGIONAL ACTION

Regional Annual Action Programme

Five collaboration units specify exchange, annual action, owner types and receipts

UNIT	ROLE	RESOURCE EXCHANGE	ANNUAL ACTION	RECEIPT
Beihai Community	neighbourhood co-test	public needs, research, devices, methods and service feedback	4 walks + 6 service improvements	issues + corrections
Future Science City	research scenario validation	public needs, research, devices, methods and service feedback	2 joint scenario challenges	challenge + test report
Huairou Science City	science-to-city transition	public needs, research, devices, methods and service feedback	2 course packs + 1 urban solution	course + feedback
Beijing Li-Town	robotics and manufacturing validation	public needs, research, devices, methods and service feedback	3 device tests + 1 supply-chain session	safety + conversion
Beijing-Tianjin-Hebei	interregional scenario network	public needs, research, devices, methods and service feedback	1 catalogue + 2 workshops + 3 off-site validations	catalogue + validation

DELIVERY RULE: enter the programme after annual agreement; register provenance, rights, budget, owner and exit.

regional action: programjoin + regional collaboration ledgerjoin

collaboration is delivered through annual agreements and receipts

11

regional action network

Human-readable drawings; authoritative data remain in JSON/GeoJSON evidence files

Make Evidence Boundaries Part of the Scheme, Not a Footnote

Sentinel green and official-facility validation are complete; formal boundaries, tenure, utilities and field survey still require specialist confirmation

15 / BASIS + MATERIALS

Planning Basis and Material-use Statement

Materials are classified, sourced, managed by use and reviewed as an updatable planning record

8

MATERIAL CLASSES

30

PLANNING FIGURES

4

PDF DELIVERABLES

2

LANGUAGE SETS

MATERIAL	SOURCE / RIGHT	PLANNING USE	MANAGEMENT REQUIREMENT	STATUS
proposal text and translations	agreed authored for this submission	original submission text	policy and standard compliance (initial approval)	REQUESTED
technical figures and diagrams	generated locally from submission data by the package build script	agreed authored graphics	promotive (geometry is labelled and cannot be created as statutory mapping)	TO PRODUCE
OpenStreetMap context	Downloaded from contributors	UK/IE 3D ambience	background context only; no ownership or statutory relevance	USED
NASA POWER climate baseline	NASA POWER already API documented and reviewed online	public weather data with attribution	context-edge (managed baseline, single writing design requires local data)	PRODUCED
Blender corridor rendering	agreed generated from 3D modelled meaning	agreed authored rendering	not an output or surveyed model	CONCEPT
generated concept scenes	Unreal Engine generated and checked for this submission	submission concept imagery	design intent only; not identifiable real-world or official representation	TO SUBMIT
Threema offline viewer	Threema B2B S1 limited access	MTI license	license review obtained license package	USED
system fonts	Microsoft Windows or available GDI system font	render only local font use	font files are not submitted	PRODUCED

STATUTORY BASIS

Produce, standardise and test sources are entered in the source register

TECHNICAL BASE

The working base is updated with official boundaries, survey and amenity-validation records

DELIVERABLE USE

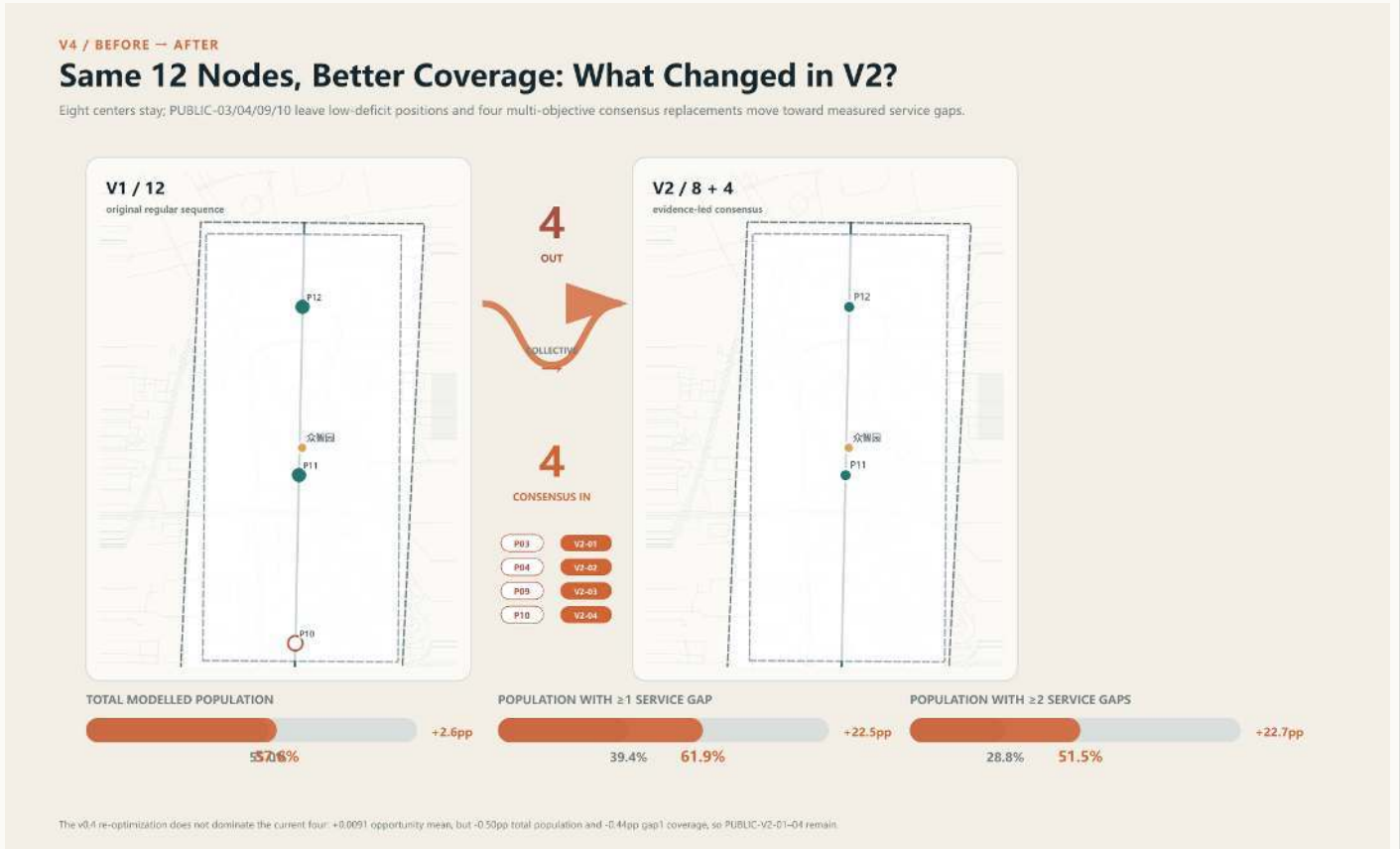
Concept valid only through planning intent; delivery proceeds through specialist design outputs

sources:gear - rights:clearance:lodgement - reports:log:right_statement:md

all materials follow source registration, use management and deliverable review

15

rights / source / clearance status



observed → response → impact → re-evaluation